

A Practical Guide to DenseNet

TOPIC -- DENSENET

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A residual neural network (also referred to as a residual network or ResNet) is a deep learning architecture in which the layers learn residual functions with reference to the layer inputs. It was developed in 2015 for image recognition, and won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) of that year. As a point of terminology, "residual connection" refers to the specific architectural motif of $x \mapsto f(x) + x$, where f is an arbitrary neural network module. The motif had been used previously (see §History for details). However, the publication of ResNet made it widely popular for feedforward networks, appearing in neural networks that are seemingly unrelated to ResNet. The residual connection stabilizes the training and convergence of deep neural networks with hundreds of layers, and is a common motif in deep neural networks, such as transformer models (e.g., BERT, and GPT models such as ChatGPT), the AlphaGo Zero system, the AlphaStar system, and the AlphaFold system.

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Previous work Residual connections were noticed in neuroanatomy, such as Lorente de No (1938). McCulloch and Pitts (1943) proposed artificial neural networks and considered those with residual connections. In 1961, Frank Rosenblatt described a three-layer multilayer perceptron (MLP) model with skip connections. The model was referred to as a "cross-coupled system", and the skip connections were forms of cross-coupled connections.

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This formulation suggests that the gradient computation of a shallower layer, $\frac{\partial E}{\partial x_{\ell}}$, always has a later term $\frac{\partial E}{\partial x_L}$ that is directly added. Even if the gradients of the $F(x_i)$ terms are small, the total gradient $\frac{\partial E}{\partial x_{\ell}}$ resists vanishing due to the added term $\frac{\partial E}{\partial x_L}$.

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Forward propagation If the output of the ℓ -th residual block is the input to the $(\ell + 1)$ -th residual block (assuming no

activation function between blocks), then the $(\ell + 1)$ -th input is:

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Basic block A basic block is the simplest building block studied in the original ResNet. This block consists of two sequential 3×3 convolutional layers and a residual connection. The input and output dimensions of both layers are equal.

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During the early days of deep learning, there were attempts to train increasingly deep models. Notable examples included the AlexNet (2012), which had 8 layers, and the VGG-19 (2014), which had 19 layers. However, stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In theory, adding additional layers to deepen a network should not result in a higher training loss, but this is what happened with VGGNet. If the extra layers can be set as identity mappings, however, then the deeper network would represent the same function as its shallower counterpart. There is some evidence that the optimizer is not able to approach identity mappings for the parameterized layers, and the benefit of residual connections was to allow identity mappings by default. In 2014, the state of the art was training deep neural networks with 20 to 30 layers. The research team for ResNet attempted to train deeper ones by empirically testing various methods for training deeper networks, until they came upon the ResNet architecture.

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Bottleneck block A bottleneck block consists of three sequential convolutional layers and a residual connection. The first layer in this block is a 1×1 convolution for dimension reduction (e.g., to $1/2$ of the input dimension); the second layer performs a 3×3 convolution; the last layer is another 1×1 convolution for dimension restoration. The models of ResNet-50, ResNet-101, and ResNet-152 are all based on bottleneck blocks.

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Stochastic depth is a regularization method that randomly drops a subset of layers and lets the signal propagate through the identity skip connections. Also known as DropPath, this regularizes training for deep models, such as vision transformers. ResNeXt (2017) combines the Inception module with ResNet. Squeeze-and-Excitation Networks (2018) added squeeze-and-excitation (SE) modules to ResNet. An SE module is applied after a convolution, and takes a tensor of shape $R \times H \times W \times C$

(height, width, channels) as input. Each channel is averaged, resulting in a vector of shape $\mathbb{R}^C$. This is then passed through a multilayer perceptron (with an architecture such as linear-ReLU-linear-sigmoid) before it is multiplied with the original tensor. It won the ILSVRC in 2017.

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where ϕ can be any activation (e.g. ReLU) or normalization (e.g. LayerNorm) operation. This design reduces the number of non-identity mappings between residual blocks, and allows an identity mapping directly from the input to the output. This design was used to train models with 200 to over 1000 layers, and was found to consistently outperform variants where the residual path is not an identity function. The pre-activation ResNet with 200 layers took 3 weeks to train for ImageNet on 8 GPUs in 2016. Since GPT-2, transformer blocks have been mostly implemented as pre-activation blocks. This is often referred to as "pre-normalization" in the literature of transformer models.

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Backward propagation The residual learning formulation provides the added benefit of mitigating the vanishing gradient problem to some extent. However, it is crucial to acknowledge that the vanishing gradient issue is not the root cause of the degradation problem, which is tackled through the use of normalization. To observe the effect of residual blocks on backpropagation, consider the partial derivative of a loss function E with respect to some residual block input x_{ℓ} . Using the equation above from forward propagation for a later residual block $L > \ell$:

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Degradation problem Sepp Hochreiter discovered the vanishing gradient problem in 1991 and argued that it explained why the then-prevalent forms of recurrent neural networks did not work for long sequences. He and Schmidhuber later designed the LSTM architecture to solve this problem, which has a "cell state" c_t that can function as a generalized residual connection. The highway network (2015) applied the idea of an LSTM unfolded in time to feedforward neural networks, resulting in the highway network. ResNet is equivalent to an open-gated highway network.